Test report 19 – Measuring

Plan 18: measurement tool in UI (browser, Tauri) and CLI with shared logic in Rust · cad-measure (new), cad-engine (`measure_parts`, `section_parts`), cad-session (`measure.rs`, protocol `Measure`), cad-cli (`measure.rs`), web UI (`app/measure.ts`, `measureUi.tsx`, viewer) · 2026-10-24 · Branch worktree-agent-ae8a7468fa02b25a2 · Commit a949494 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

Measurement kinds	Analytic cases (CLI)	cargo test --workspace	clippy · fmt	Vitest · Playwright	Tauri T17
7	**21 / 21**	**574 / 574**	**clean · clean**	**233 · 131 / 131**	**passed**
distance, radius, angle, length, area, body, wall thickness – also in the section	within 10^{-6} relative (kernel volume 10^{-3})	22 of them new	-D warnings	new: M01–M11	6 checks, native kernel

Response under the mouse
2.9 ms
median (browser, Wasm, 42 responses)

1 Task and result

Task: extend the measurement tool in the UI (browser and Tauri) and in the CLI – distance point–point and between points, edges, faces (minimal, with foot points, parallel planes as normal distance), radius/diameter of circular edges, arcs, cylinder, cone, sphere and torus faces exactly from the B-rep with axis/center, angle, length, area, bounding box/volume; measuring in the active section (snap to contours, distances in the plane, radius of section circles, wall thickness, contour area) with exact section curves; snap, dimension lines and labels, multiple measurements, delete, copy as text/comment/`cad measure`; link to the language; the same measurement kinds in `cad measure` with JSON. Implemented:

- **Shared logic in Rust:** new crate `cad-measure` (exact edges and faces, distance, angle, radius, area, section, wall thickness), in `cad-session` the measurement model per node, snap, provenance, texts and CLI arguments – the same functions serve the protocol (UI) and the CLI references.
- **Exactness:** analytic geometry in closed form; section curves as plane ∩ B-rep from the kernel, their segments on analytic surfaces as line, circle or ellipse (with the deviation of the kernel curve); otherwise with a stated tolerance. Every measurement carries `exact` and `tol`.
- **UI:** mode "Measure" (German UI: „Messen“) with context bar (measurement kind, "In section"), snap with indicator, dimension lines, guide lines, labels (checked light/dark), list with copy and delete – as standalone building blocks for the UI rework.
- **CLI:** `--distance`, `--radius`, `--angle`, `--length`, `--area face@P`, `--body`, `--section` … (contours, circles, thinnest wall), `--wall`, `--at`; element references `vertex@P`, `mid@P`, `center@P`, `edge@P`, `face@P`, `body@P`, `region@P`; JSON `measurements` (protocol type), `section`, `at`.

2 Decisions

Question	Decision
Where measuring happens	In the engine (worker or engine process) on the neutral B-rep of the node parts in context coordinates; the UI sends ray, its raycast hit (node, face ID, point), pixel size and snap radius, and receives the element under the mouse and, after the last click, the measurement with dimension lines. As with the sketch editor, no selection state in the engine.

Distance	Coarse search over the tessellation (BVH), refinement by alternating projection onto the exact curves and surfaces; outside a trimmed face onto its boundary. Parallel planes: normal distance as the value, the minimum distance of the bounded faces alongside; parallel axes: axis distance.
Radius	From the B-rep parameters; fillets on circular edges (in SMLib surfaces of revolution with a circular-arc generatrix) are recognized, free-form via principal curvature (approximated).
Angle of two planes	Angle between the planes (0–90°, chamfer 45°), plus the opening angle at the intersection line and the angle of the normals.
Section	Kernel Boolean with the kept half-space (convention of `cad render --section`), cap faces as regions; segments from the cut surface in closed form. One Boolean per part and plane, cached.
Wall thickness	At a contour point perpendicular to the contour into the material up to the next contour; without a point the minimum over samples (estimate).
Link to the language	Face tags → nodes → lines and named instances; plus grid points, grid lines, grid planes and anchors on which a point, an axis or a plane lies.
CLI references	The element nearest to a point (`face@[x,y,z]`): unambiguous and readable, the UI copies measurements in this form; `--at P` shows what lies at P.
Place in the UI	Context bar and list as standalone components (`MeasureBar`, `MeasurePanel`); for now in the stack at the top right and to the right of the view, per the UI concept later at the top left below the toolbar and as a tab "Measure".

3 Flow

```
Mouse moved  → Measure { kind, picks, ray, hit {node, face, point}, pixel, snap_px, free, section? | clip? }
             ← Measure { hover {kind, label, point, reference, origin, lines}, prompt, section {regions, area,
exact} }
Click        → Measure { …, click: true }
             ← picks (selection stored)  or  result: Measurement { value, label, values, lines, guides,
points,
                                                                  at, details, exact, tol, text, cli }
Engine:  measure_model(node)  = per part B-rep + tessellation (context)      measure_section(node, plane) =
Boolean
         snap → vertex | edge midpoint | circle center | edge | face  (in section: contour, region)
         measurement → cad_measure::{distance, radius, angle, face_area, Section::wall …}
CLI:     face@[x,y,z] → same selection → same measurement → JSON/text
```

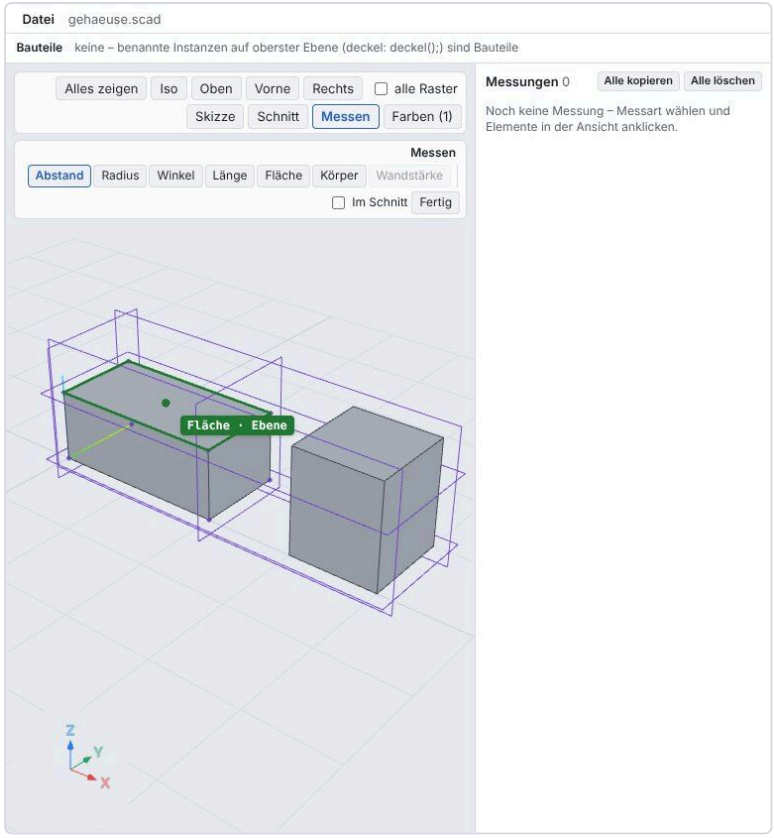
4 Measurement kinds in the UI

All scenarios click with the mouse (Playwright, Chromium with SwiftShader) and check the values against the geometry.

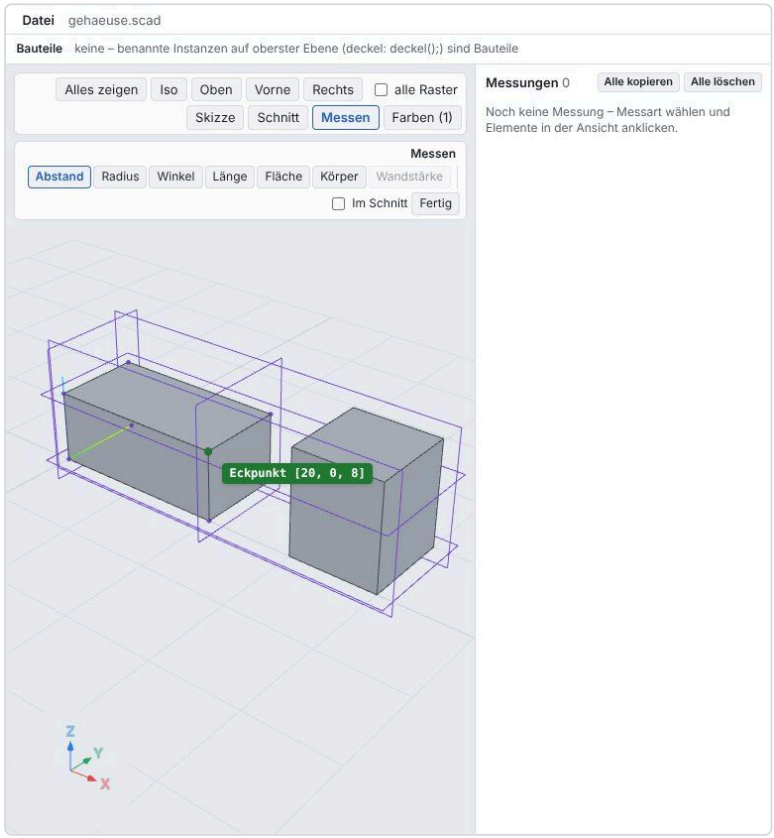
M01 Measure mode on/off, snap

"Measure" in the view toolbar turns measure mode on: the context bar shows the measurement kinds (distance, radius, angle first as in the UI concept, plus length, area, body, wall thickness in section) and "In section". Under the mouse the engine snaps to vertices, edge midpoints, circle centers, edges and faces (snap radius from the settings); label and status line name the element and where it comes from. Esc exits the mode.

```
grid g {
  x: a = 0, b = 20;
  y: v = 0, h = 10;
  z: u = 0, o = 8;
}
box(g[a, v, u], g[b, h, o]);
translate([30, 0, 0]) cube([10, 10, 12]);
```



mouse over the top face: face highlighted, label, provenance g.o



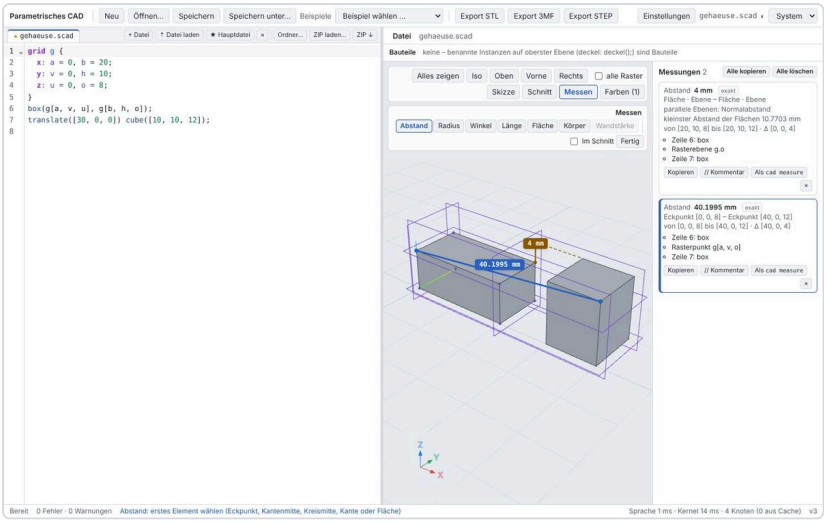
near a corner: snap to the vertex (grid point g[b, v, o])

- hover face with grid plane g.o
- vertex [20, 0, 8] = grid point g[b, v, o]
- edge midpoint
- Esc: mode off

M02 Distance

Two clicks: minimum distance of the elements with foot points, from the exact B-rep (alternating projection onto planes, cylinders, … in closed form). For parallel planes the normal distance applies. The dimension line connects the foot points, the label sits in the middle; the list on the right shows values, Δx/Δy/Δz and origin.

```
grid g {
  x: a = 0, b = 20;
  y: v = 0, h = 10;
  z: u = 0, o = 8;
}
box(g[a, v, u], g[b, h, o]);
translate([30, 0, 0]) cube([10, 10, 12]);
```



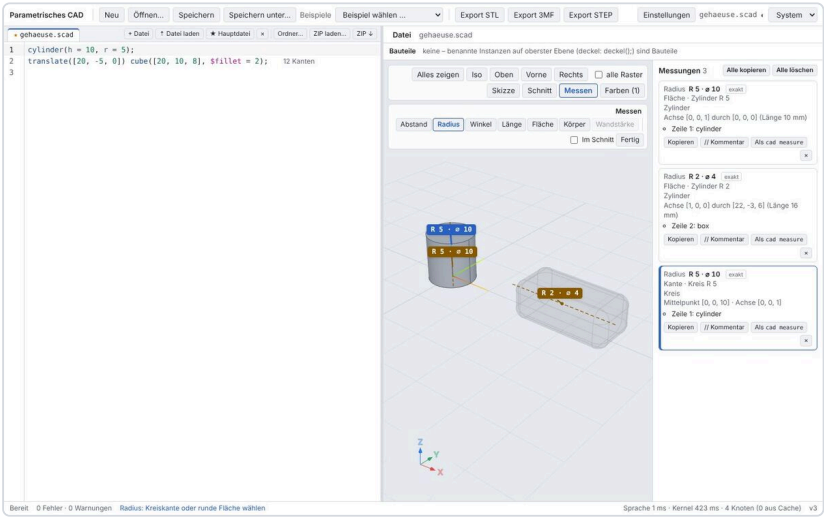
two measurements: normal distance of the top faces 4 mm, vertex–vertex with Δ

- normal distance 4 mm (exact)
- vertices 40.199502 mm, Δx 40, Δz 4
- cad measure --distance vertex@[0,0,8] vertex@[40,0,12]

M03 Radius and diameter

Radius of a circular edge or round face from the B-rep: cylinder, cone, sphere, torus in closed form, fillets on circular edges (surface of revolution of a circular arc) recognized, free-form via curvature. Radius line, center and axis are drawn.

```
cylinder(h = 10, r = 5);
translate([20, -5, 0]) cube([20, 10, 8],
$fillet = 2);
```

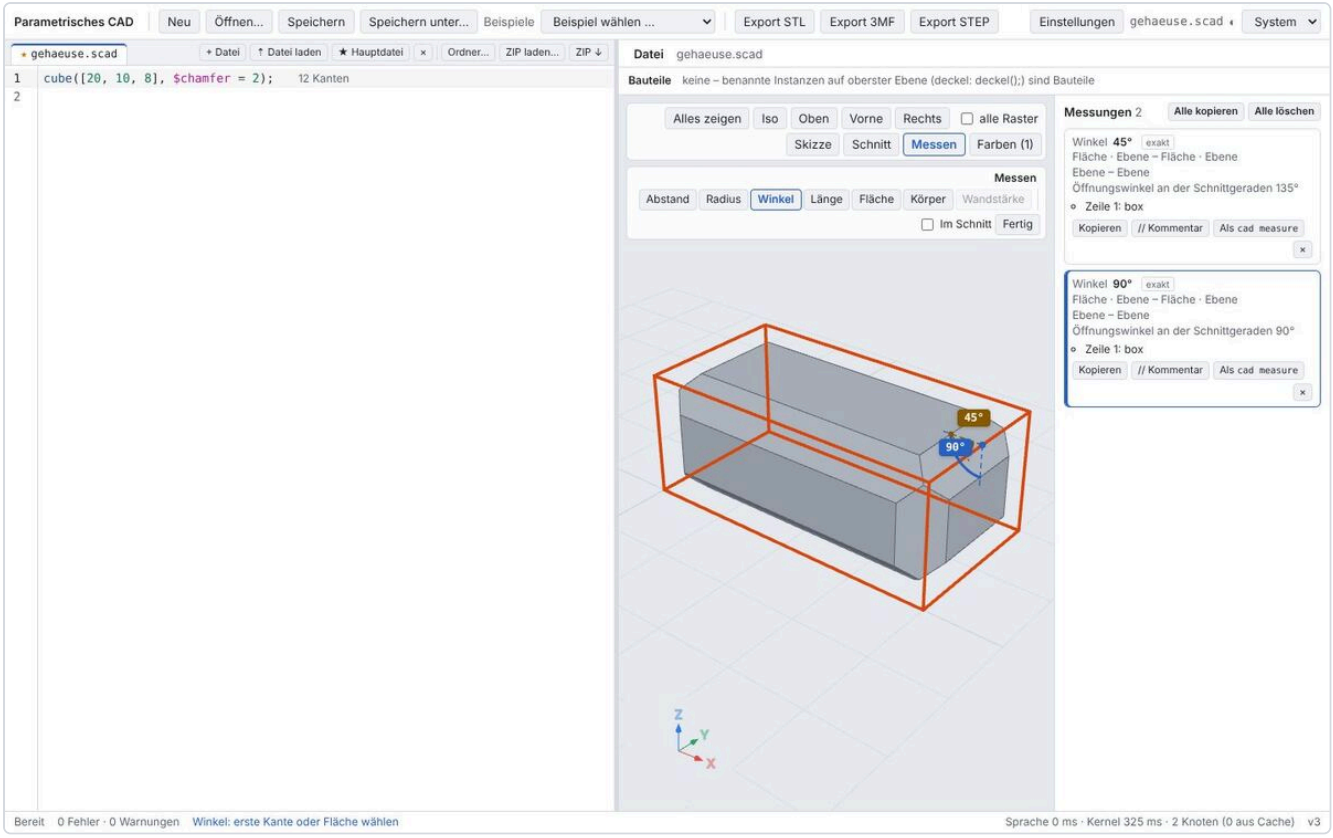


radius of the cylinder (axis dashed), of the fillet (R 2) and of the circular edge

- cylinder R 5 · ⌀ 10 with axis
- fillet R 2
- circular edge R 5

M04 Angle

Angle between two edges or faces: planes via their normals (shown: the angle between the planes, plus the opening angle at the intersection line), cylinders and cones via their axis, edges at a common vertex pointing away from the vertex. The arc sits at the intersection line.



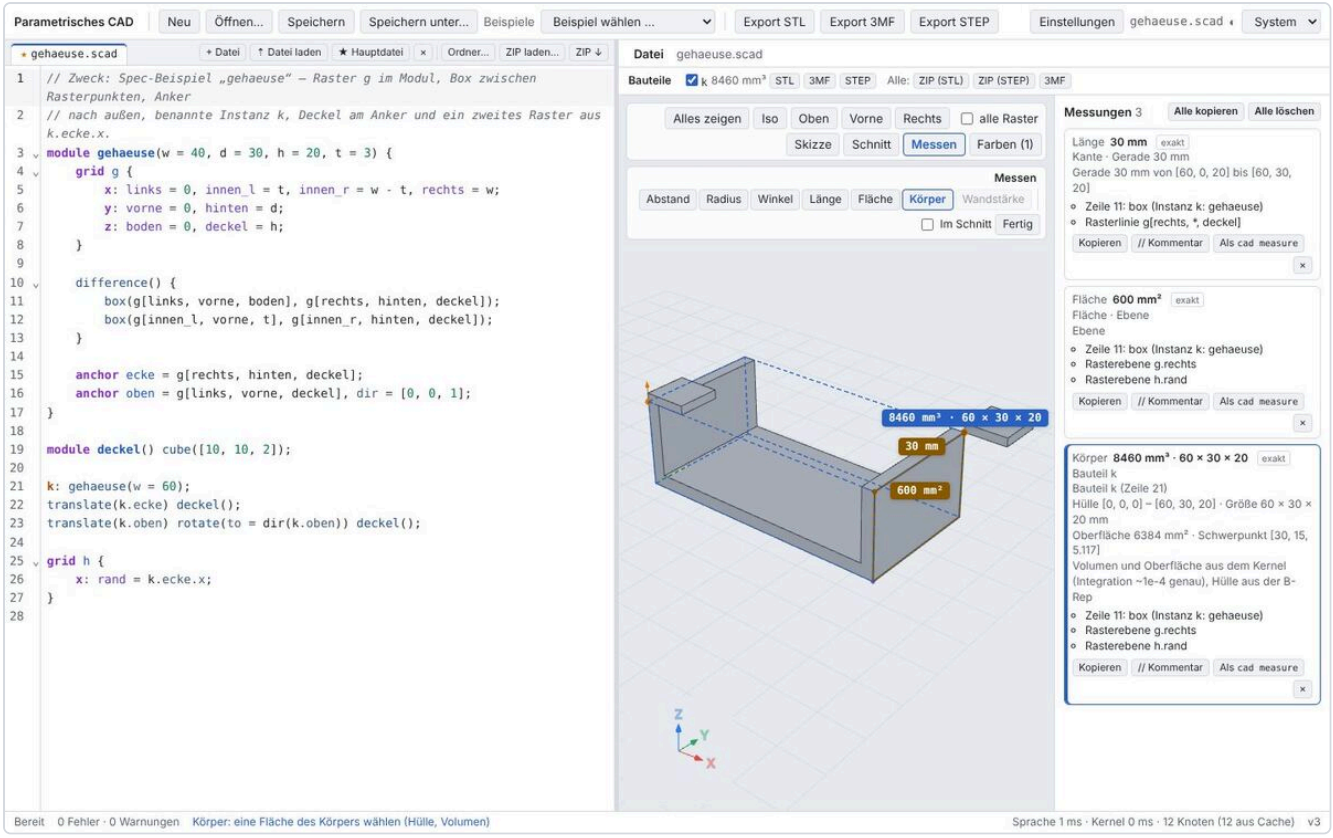
chamfer 45° (opening angle 135°) and top face–side 90°, with angle arc

```
cube([20, 10, 8], $chamfer = 2);
```

- chamfer 45°
- opening angle 135°
- top face–side 90°

M05 Length, area, body

Length of an edge (exact for lines and arcs, otherwise Gauss–Legendre), area of a face (planar exactly via the boundary, curved via projected subdivision), body: the part of the chosen face (named instance k) with volume and surface area from the kernel and the bounding box from the B-rep (dashed box).



edge length 30 mm, area 600 mm², part k: volume and bounding box 60 × 30 × 20

`examples/spec/gehaeuse.scad`

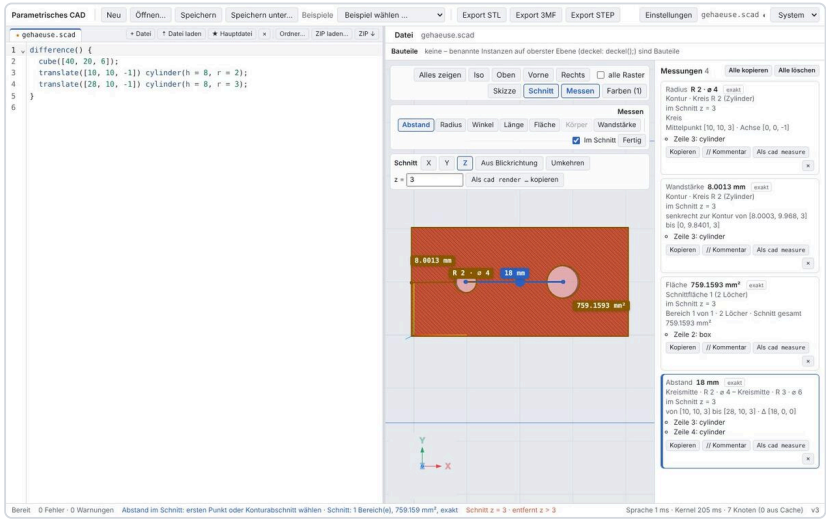
- length 30 mm
- area 600 mm²
- part k: 8460 mm³, 60 × 30 × 20

5 Measuring in the section

M06 Measuring in the section

"In section" measures in the active section plane: the contours come from the kernel (plane ∩ B-rep via a Boolean with the kept half-space), the segments on cylinders, cones, spheres and tori in closed form (here exact circles). Snap to vertices, midpoints and circle centers of the contour; wall thickness across the contour; area of a region (contour minus holes).

```
difference() {
  cube([40, 20, 6]);
  translate([10, 10, -1]) cylinder(h = 8,
r = 2);
  translate([28, 10, -1]) cylinder(h = 8,
r = 3);
}
```



section z = 3: radius of the bore, wall thickness 8 mm, section area, distance of the circle centers 18 mm

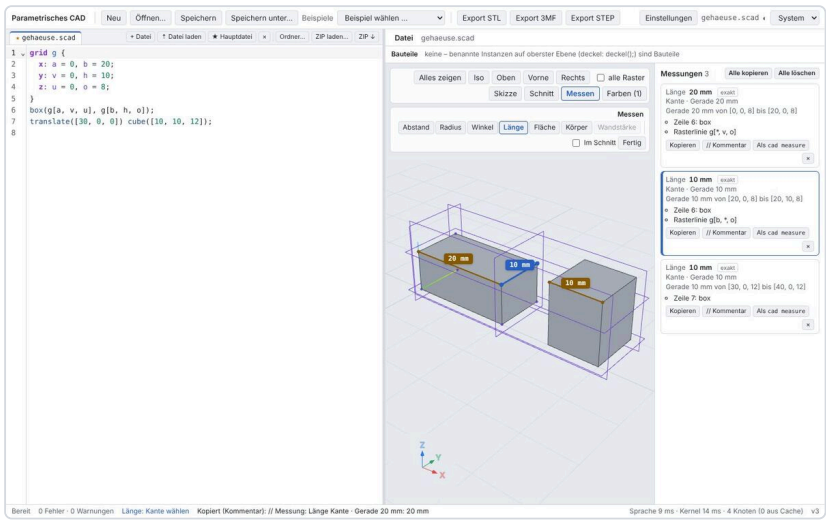
- section exact, 1 region
- bore R 2 · ⌀ 4
- wall thickness 8 mm
- area 759.1593 mm²
- circle centers 18 mm

6 List, copy, link to the language, theme

M07 List of measurements

Multiple measurements stay visible at the same time (also after exiting the mode). A click on a label or entry selects a measurement, Del or × deletes it, "Delete all" deletes all. Each measurement can be copied as text, as a code comment (`// Messung: …`) and as a `cad measure` call (like "Copy as cad render" for the section).

```
grid g {
  x: a = 0, b = 20;
  y: v = 0, h = 10;
  z: u = 0, o = 8;
}
box(g[a, v, u], g[b, h, o]);
translate([30, 0, 0]) cube([10, 10, 12]);
```

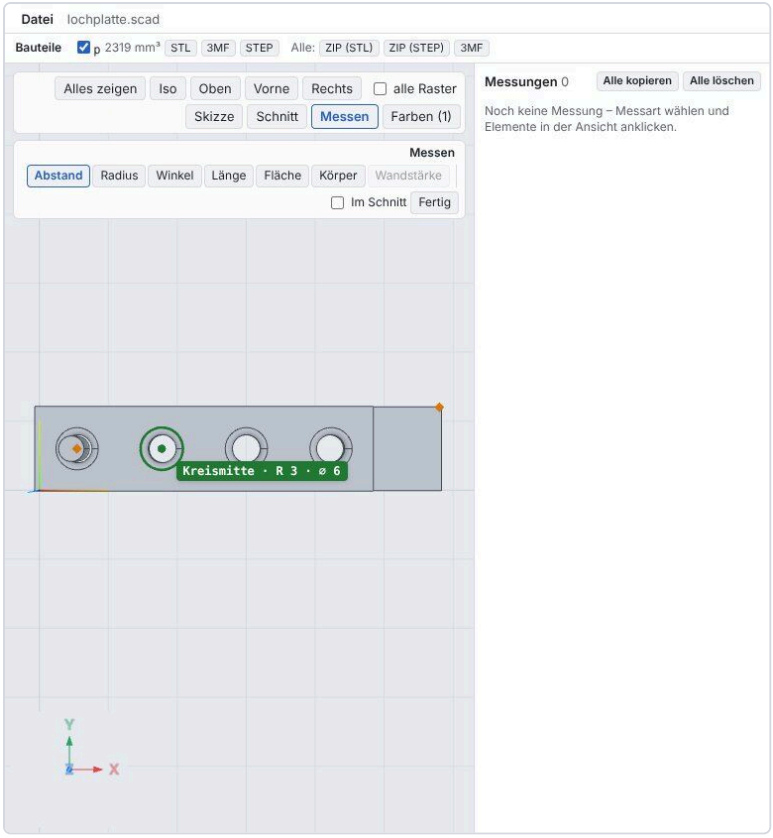


three lengths, one selected (accent color in view and list)

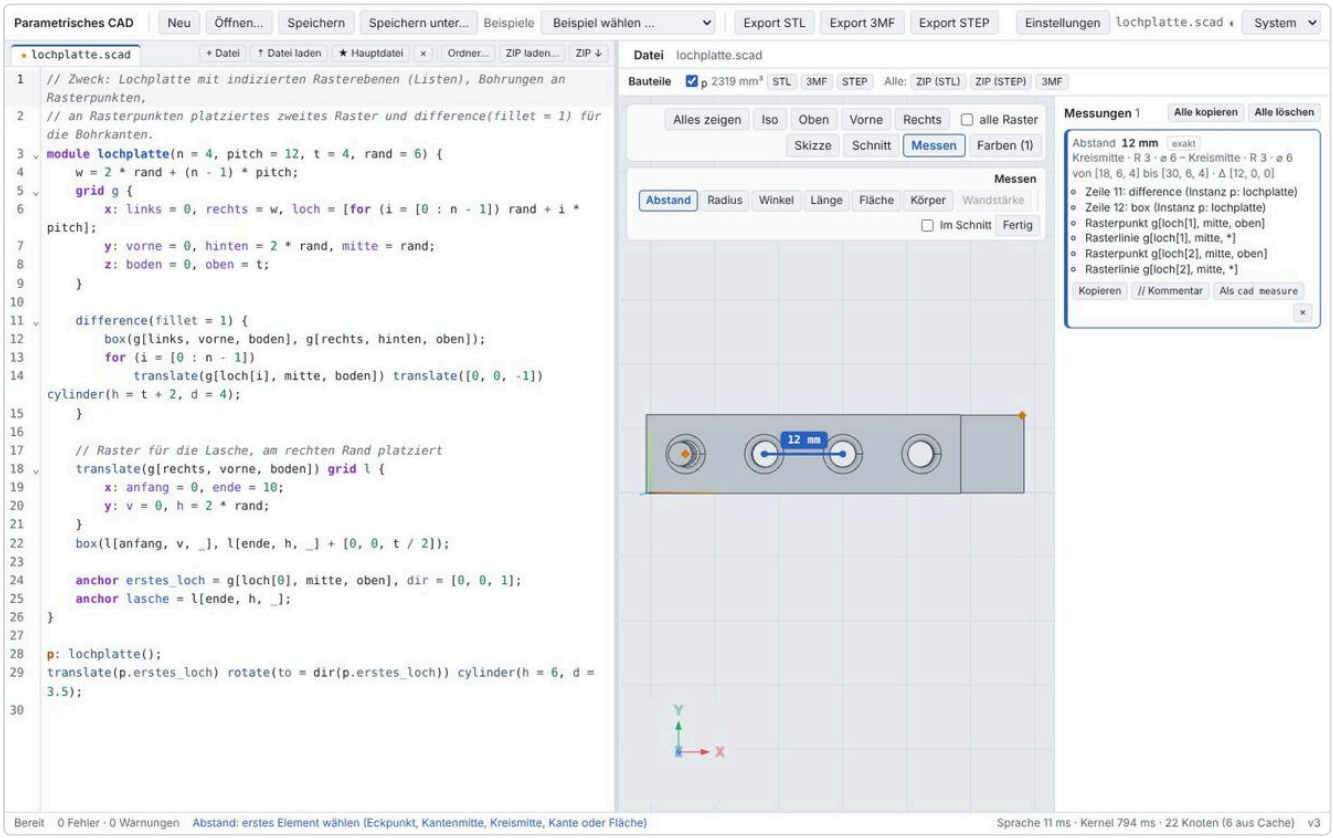
- copied: cad measure gehaeuse.scad --length 'edge@[10.049762,0,8]'
- comment // Messung: Länge …
- Del deletes the selected one
- Delete all

M08 Link to the language

For each chosen element the engine names the code lines of the faces (tags → nodes), the named instance (p: lochplatte) and grids or anchors on which it lies: grid points and grid planes, grid lines for axes (g[loch[0], mitte, *]), anchors (p.erstes_loch).



circle center of a bore (chamfer/fillet) with origin



distance of the circle centers = pitch 12 mm, origin in the list

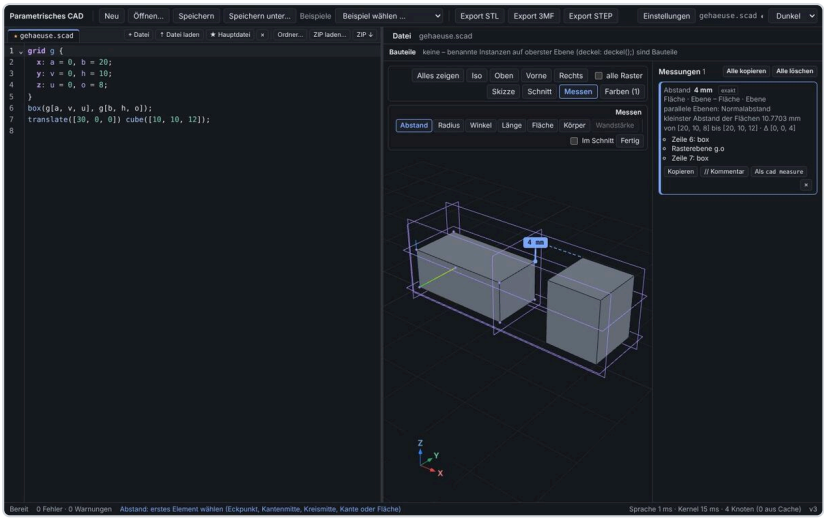
`examples/spec/lochplatte.scad`

- circle center with instance p
- distance 12 mm = pitch

M09 Dark theme

Dimension lines and labels use the tokens `--view-measure`/`--on-measure` (light: dark gold with white text, dark: gold with dark text); contrast.test.ts checks text ≥ 4.5:1 and line against background ≥ 3:1.

```
grid g {
  x: a = 0, b = 20;
  y: v = 0, h = 10;
  z: u = 0, o = 8;
}
box(g[a, v, u], g[b, h, o]);
translate([30, 0, 0]) cube([10, 10, 12]);
```



measurement in the dark theme

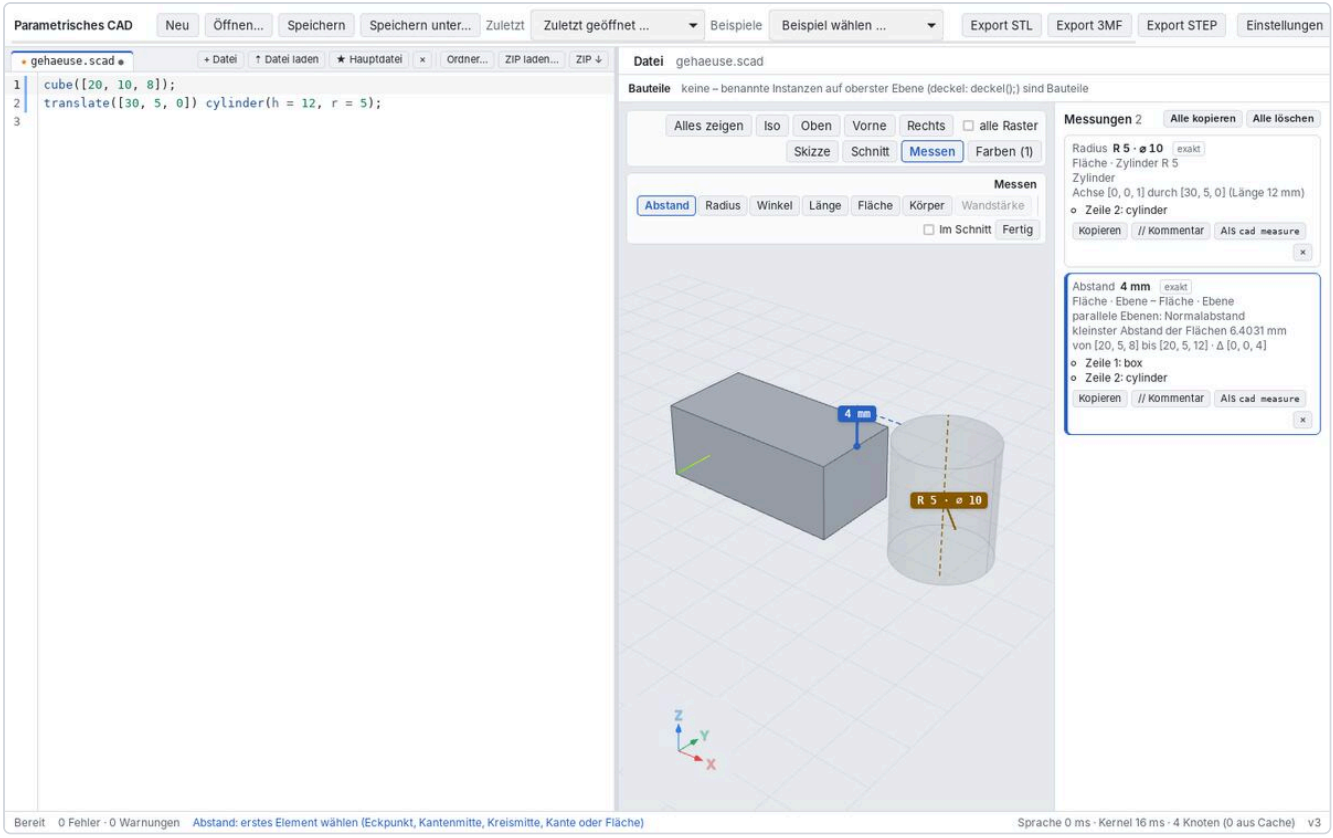
- label contrast 8.10:1

7 Desktop app (Tauri)

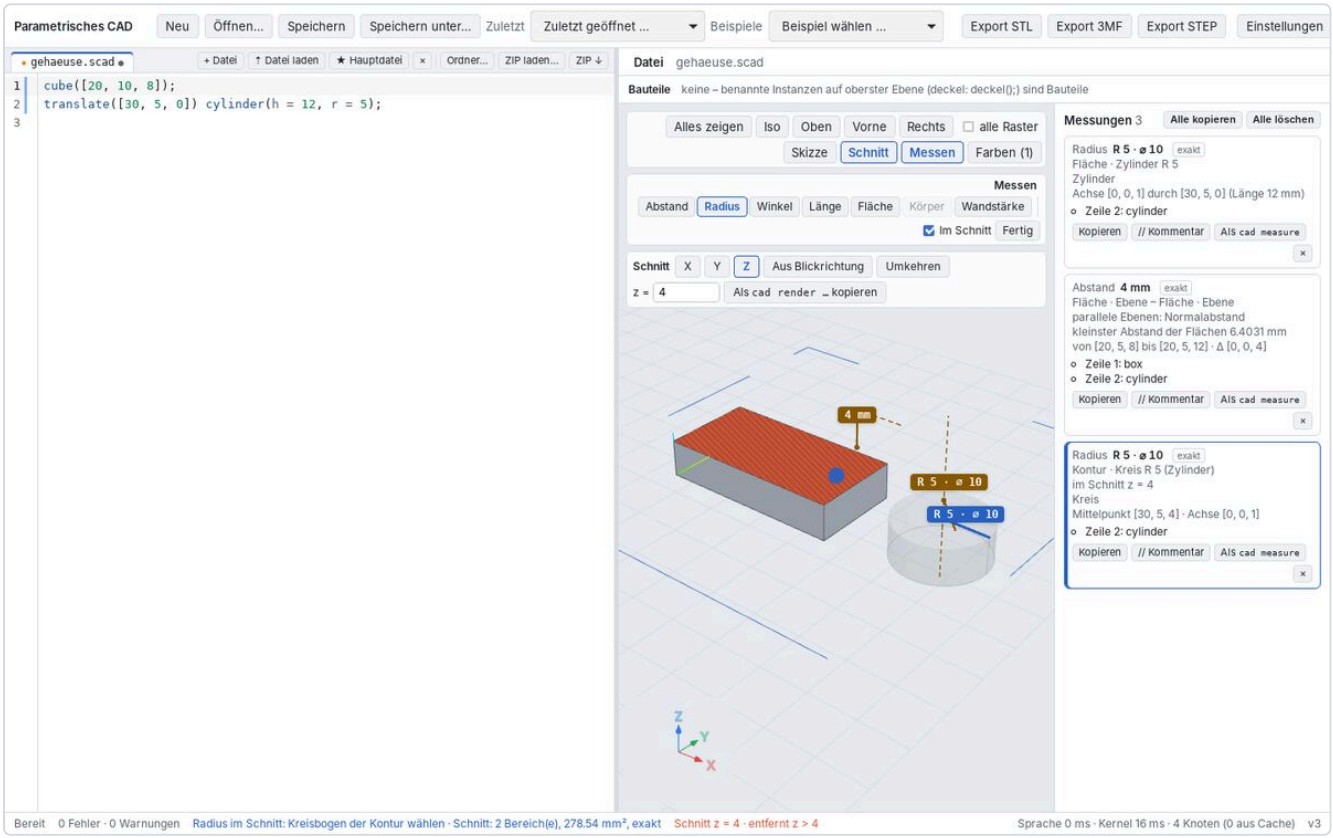
Scenario T17 in `web/e2e-tauri/desktop.e2e.mjs` (tauri-driver, WebKitWebDriver, Xvfb): the app `cad` with the native engine process, messages via `TauriTransport`; the view is operated with pointer events on the canvas.

Result: **passed** in 6.1 s, response under the mouse median 11.0 ms.

- radius of the cylinder: R 5 · ⌀ 10
- normal distance of the top faces: 4 mm (exact)
- label in the view: 4 mm
- section exact: 2 regions, 278.5398 mm²
- circle in the section: R 5 · ⌀ 10
- measure mode exited



radius of the cylinder and normal distance of the top faces



radius of the section circle (section z = 4)

8 CLI: `cad measure`

The same measurements without UI, for agents with `--json`. Output from `tools/report/report19-data.sh` (release binary, without daemon; times per call as the mean of 3 runs including loading and kernel).

Housing in section z = 10: regions, wall thickness, inner dimension (133 ms)

```
$ cad measure examples/spec/gehaeuse.scad --section
z=10 --wall \[0\,15\,10\] --distance edge@\
[3\,15\,10\] edge@\[57\,15\,10\]
Schnitt z = 10 (entfernt z > 10): 2 Bereich(e),
Fläche 180 mm² (exakt: Konturen aus Geraden und
Kreisen der B-Rep)
  Bereich 1: 90 mm², Hülle [57, 0, 10] – [60, 30,
10], 0 Loch/Löcher
  Bereich 2: 90 mm², Hülle [0, 0, 10] – [3, 30, 10],
0 Loch/Löcher
  kleinste Wandstärke ≈ 3 mm von [57, 28.125, 10] bis
[60, 28.125, 10] (8 Stellen je Konturabschnitt;
genau: --wall P)
Abstand Kontur · Gerade 30 mm (Ebene) – Kontur ·
Gerade 30 mm (Ebene): 54 mm
  von [3, 0, 10] bis [57, 0, 10] · Δ [54, 0, 0]
  Kontur: Zeile 12: box (Instanz k: gehaeuse)
  Kontur: Zeile 12: box (Instanz k: gehaeuse)
  im Schnitt --section z=10
Wandstärke Punkt [0,15,10]: 3 mm
  senkrecht zur Kontur von [0, 15, 10] bis [3, 15,
10]
  im Schnitt --section z=10
```



`cad render gehaeuse.scad --section z=10 --view top`: the two wall regions

Housing: inner distance, part, edge length (114 ms)

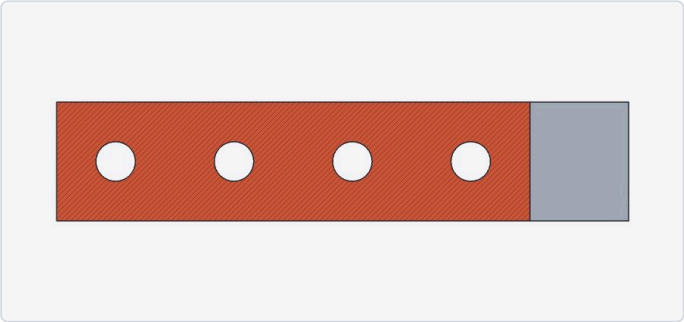
```
$ cad measure examples/spec/gehaeuse.scad --distance face@\[3\,15\,10\] face@\[57\,15\,10\] --body face@\
[0\,15\,10\] --length edge@\[60\,15\,20\]
Abstand Fläche · Ebene – Fläche · Ebene: 54 mm
  parallele Ebenen: Normalabstand
  von [3, 30, 20] bis [57, 30, 20] · Δ [54, 0, 0]
  Fläche: Zeile 12: box (Instanz k: gehaeuse)
  Fläche: Rasterebene g.innen_l
  Fläche: Zeile 12: box (Instanz k: gehaeuse)
  Fläche: Rasterebene g.innen_r
Länge Kante · Gerade 30 mm: 30 mm
  Gerade 30 mm von [60, 0, 20] bis [60, 30, 20]
  Kante: Zeile 11: box (Instanz k: gehaeuse)
  Kante: Rasterlinie g[rechts, *, deckel]
Körper Bauteil k: 8460 mm³
  Bauteil k (Zeile 21)
  Hülle [0, 0, 0] – [60, 30, 20] · Größe 60 × 30 × 20 mm
  Oberfläche 6384 mm² · Schwerpunkt [30, 15, 5.117]
  Volumen und Oberfläche aus dem Kernel (Integration ~1e-4 genau), Hülle aus der B-Rep
  Körper: Zeile 11: box (Instanz k: gehaeuse)
  Körper: Rasterebene g.links
```

Hole plate: bore, fillet, pitch (1145 ms)

```
$ cad measure examples/spec/lochplatte.scad --radius face@\[8\,6\,2\] --radius face@\[8.292893\,6\,3.707107\] --distance center@\[6\,6\,0\] center@\[18\,6\,0\]
Abstand Kreismitte · R 3 · ⌀ 6 – Kreismitte · R 3 · ⌀ 6: 12 mm
  von [6, 6, 0] bis [18, 6, 0] · Δ [12, 0, 0]
  Kreismitte: Zeile 11: difference (Instanz p: lochplatte)
  Kreismitte: Zeile 12: box (Instanz p: lochplatte)
  Kreismitte: Rasterpunkt g[loch[0], mitte, boden]
  Kreismitte: Rasterlinie g[loch[0], mitte, *]
  Kreismitte: Zeile 11: difference (Instanz p: lochplatte)
  Kreismitte: Zeile 12: box (Instanz p: lochplatte)
  Kreismitte: Rasterpunkt g[loch[1], mitte, boden]
  Kreismitte: Rasterlinie g[loch[1], mitte, *]
Radius Fläche · Zylinder R 2: 2 mm
  Zylinder
  Achse [0, 0, 1] durch [6, 6, 1] (Länge 2 mm)
  Fläche: Zeile 14: cylinder (Instanz p: lochplatte)
  Fläche: Rasterlinie g[loch[0], mitte, *]
Radius Fläche · Rotationsfläche: 1 mm
  Rotationsfläche (Kreisbogen)
  Mittelpunkt [6, 6, 3] · Achse [0, 0, 1] · Ringradius 3 mm
  Fläche: Zeile 11: difference (Instanz p: lochplatte)
  Fläche: Rasterlinie g[loch[0], mitte, *]
```

Hole plate in section z = 2: circles, area, radius, wall (1234 ms)

```
$ cad measure examples/spec/lochplatte.scad --section z=2 --radius edge@\[8\,6\,2\] --wall \[8\,6\,2\]
Schnitt z = 2 (entfernt z > 2): 1 Bereich(e), Fläche 645.7345 mm² (exakt: Konturen aus Geraden und Kreisen der B-Rep)
  Bereich 1: 645.7345 mm², Hülle [0, 0, 2] – [58, 12, 2], 4 Loch/Löcher
  Kreis R 2 · ⌀ 4 bei [6, 6, 2] (center@[6,6,2])
  Kreis R 2 · ⌀ 4 bei [18, 6, 2] (center@[18,6,2])
  Kreis R 2 · ⌀ 4 bei [30, 6, 2] (center@[30,6,2])
  Kreis R 2 · ⌀ 4 bei [42, 6, 2] (center@[42,6,2])
  kleinste Wandstärke ≈ 4.0039 mm von [18.125, 12, 2] bis [18.125, 7.9961, 2] (8 Stellen je Konturabschnitt; genau: --wall P)
Radius Kontur · Kreis R 2 (Zylinder): 2 mm
  Kreis
  Mittelpunkt [6, 6, 2] · Achse [0, 0, -1]
  Kontur: Zeile 14: cylinder (Instanz p: lochplatte)
  Kontur: Rasterlinie g[loch[0], mitte, *]
  im Schnitt --section z=2
Wandstärke Punkt [8,6,2]: 8 mm
  senkrecht zur Kontur von [8, 6, 2] bis [16, 6, 2]
  im Schnitt --section z=2
```



cad render lochplatte.scad --section z=2 --view top

What lies at P: `--at` (1136 ms)

```
$ cad measure examples/spec/lochplatte.scad --at \[8\,6\,2\]
Bei [8, 6, 2]:
  vertex@[8,6,1]              Eckpunkt [8, 6, 1] (1 mm entfernt)
                              Zeile 14: cylinder (Instanz p: lochplatte)
                              Zeile 11: difference (Instanz p: lochplatte)
  mid@[8,6,2]                 Kantenmitte · Gerade 2 mm (0 mm entfernt)
                              Zeile 14: cylinder (Instanz p: lochplatte)
  center@[9,6,1]              Kreismitte · R 1 · ⌀ 2 (1.4142 mm entfernt)
                              Zeile 11: difference (Instanz p: lochplatte)
  edge@[8,6,2]                Kante · Gerade 2 mm (0 mm entfernt)
                              Zeile 14: cylinder (Instanz p: lochplatte)
  face@[8,6,2]                Fläche · Zylinder R 2 (0 mm entfernt)
                              Zeile 14: cylinder (Instanz p: lochplatte)
                              Rasterlinie g[loch[0], mitte, *]
```

Angle of a chamfer (494 ms)

```
$ cad measure examples/spec/fase_chamfer.scad --angle face@\[39.5\,12\,9.5\] face@\[20\,12\,10\] --angle face@\
[20\,12\,10\] face@\[40\,12\,5\]
Winkel Fläche · Ebene — Fläche · Ebene: 45°
  Ebene — Ebene
  Öffnungswinkel an der Schnittgeraden 135°
  Fläche: Zeile 9: chamfer
  Fläche: Zeile 10: box
  Fläche: Rasterebene g.o
Winkel Fläche · Ebene — Fläche · Ebene: 90°
  Ebene — Ebene
  Öffnungswinkel an der Schnittgeraden 90°
  Fläche: Zeile 10: box
  Fläche: Rasterebene g.o
  Fläche: Zeile 10: box
  Fläche: Rasterebene g.b
```

JSON for agents (excerpt: radius of the bore)


```
{
 "kind": "distance",
 "title": "Abstand",
 "value": 12,
 "unit": "mm",
 "label": "12 mm",
 "values": [
  {
   "key": "distance",
   "label": "Abstand",
   "unit": "mm",
   "value": 12
  },
  {
   "key": "dx",
   "label": "Δx",
   "unit": "mm",
   "value": 12
  },
  {
   "key": "dy",
   "label": "Δy",
   "unit": "mm",
   "value": 0
  },
  {
   "key": "dz",
   "label": "Δz",
   "unit": "mm",
   "value": 0
  }
 ],
 "exact": true,
 "details": [
  "von [6, 6, 0] bis [18, 6, 0] · Δ [12, 0, 0]"
 ],
 "picks": [
  {
   "kind": "center",
   "reference": "center@[6,6,0]",
   "label": "Kreismitte · R 3 · ⌀ 6",
   "origin": [
    "Zeile 11: difference (Instanz p: lochplatte)",
    "Zeile 12: box (Instanz p: lochplatte)",
    "Rasterpunkt g[loch[0], mitte, boden]",
    "Rasterlinie g[loch[0], mitte, *]"
   ]
  },
  {
   "kind": "center",
   "reference": "center@[18,6,0]",
   "label": "Kreismitte · R 3 · ⌀ 6",
   "origin": [
    "Zeile 11: difference (Instanz p: lochplatte)",
    "Zeile 12: box (Instanz p: lochplatte)",
    "Rasterpunkt g[loch[1], mitte, boden]",
    "Rasterlinie g[loch[1], mitte, *]"
   ]
  }
 ],
 "cli": [
  "--distance",
  "center@[6,6,0]",
  "center@[18,6,0]"
 ],
 "text": "Abstand Kreismitte · R 3 · ⌀ 6 – Kreismitte · R 3 · ⌀ 6: 12 mm\n  von [6, 6, 0] bis [18, 6, 0] · Δ
```

```
[12, 0, 0]\n  Kreismitte: Zeile 11: difference (Instanz p: lochplatte)\n  Kreismitte: Zeile 12: box (Instanz p:
lochplatte)\n  Kreismitte: Rasterpunkt g[loch[0], mitte, boden]\n  Kreismitte: Rasterlinie g[loch[0], mitte,
*]\n  Kreismitte: Zeile 11: difference (Instanz p: lochplatte)\n  Kreismitte: Zeile 12: box (Instanz p:
lochplatte)\n  Kreismitte: Rasterpunkt g[loch[1], mitte, boden]\n  Kreismitte: Rasterlinie g[loch[1], mitte,
*]"
}
```

JSON of the section (excerpt)


```
{
 "plane": {
  "normal": [
   0,
   0,
   1
  ],
  "point": [
   0,
   0,
   2
  ]
 },
 "exact": true,
 "max_dev": 0,
 "area": 645.734517543,
 "regions": [
  {
   "area": 645.734517543,
   "holes": 4,
   "bounds": {
    "max": [
     58,
     12,
     2
    ],
    "min": [
     0,
     0,
     2
    ]
   },
   "contours": 5
  }
 ],
 "circles": [
  {
   "center": [
    6,
    6,
    2
   ],
   "diameter": 4,
   "full": true,
   "radius": 2
  },
  {
   "center": [
    18,
    6,
    2
   ],
   "diameter": 4,
   "full": true,
   "radius": 2
  },
  {
   "center": [
    30,
    6,
    2
   ],
   "diameter": 4,
   "full": true,
   "radius": 2
  },
```

```
  {
   "center": [
    42,
    6,
    2
   ],
   "diameter": 4,
   "full": true,
   "radius": 2
  }
 ],
 "min_wall": {
  "from": [
   18.125,
   12,
   2
  ],
  "samples_per_segment": 8,
  "thickness": 4.00391007217,
  "to": [
   18.125,
   7.99608992783,
   2
  ]
 }
}
```


9 Analytic check

Small models, measured via the CLI (same logic as the UI), against the formula. Limit 10^{-6} relative; volume comes from the kernel's integration (SMLib, ~10^{-4}) and has the limit 10^{-3}.

Case	Call	expected	measured	deviation	exact
Cylinder r 5, radius	`--radius face@[5,0,5]`	5.000000	5.000000	0	yes
Cylinder r 5, lateral area	`--area face@[5,0,5]`	314.159265	314.159265	-3.0e-7	yes
Cylinder r 5, circular edge length	`--length edge@[0,5,10]`	31.415927	31.415927	2.1e-12	yes
Cylinder, cap faces normal distance	`--distance face@[1,1,10] face@[1,1,0]`	10.000000	10.000000	0	yes
Cone r 4→2, radius at h 5	`--radius face@[3,0,5]`	3.000000	3.000000	0	yes
Cone r 4→2, half angle	`--radius face@[3,0,5]`	11.309932	11.309932	-2.0e-11	yes
Cone r 4→2, lateral area	`--area face@[3,0,5]`	192.228507	192.228425	-8.2e-5	yes
Sphere r 5, radius	`--radius face@[0,0,5]`	5.000000	5.000000	0	yes
Sphere r 5, volume (kernel)	`--body body@[0,0,5]`	523.598776	523.598364	-4.1e-4	yes
Torus R 10 r 2, tube radius	`--radius face@[12,0,0]`	2.000000	2.000000	0	yes
Torus R 10 r 2, ring radius	`--radius face@[12,0,0]`	10.000000	10.000000	0	yes
Fillet r 2 (straight edge)	`--radius face@[19.414214,5,7.414214]`	2.000000	2.000000	0	yes
Fillet r 1 (circular edge)	`--radius face@[12.292893,10,5.292893]`	1.000000	1.000000	0	yes
Chamfer 2 mm, angle to top face	`--angle face@[19,5,7] face@[10,5,8]`	45.000000	45.000000	0	yes
Chamfer 2 mm, opening angle	`--angle face@[19,5,7] face@[10,5,8]`	135.000000	135.000000	0	yes
Section through bore r 3, radius	`--section z=3 --radius edge@[13,10,3]`	3.000000	3.000000	0	yes

Section, area of the region	`--section z=3 --area region@[1,1,3]`	371.725666	371.725666	3.1e-10	yes
Section, wall thickness bore–edge	`--section z=3 --wall [7,10,3]`	7.000000	7.000000	0	yes
Oblique section (45°) through bore r 3: semi-major axis	`--section-plane 10,10,5,1,0,1 --radius edge@[10,13,5]`	4.242641	4.242641	7.1e-13	yes
Oblique section (45°): semi-minor axis	`--section-plane 10,10,5,1,0,1 --radius edge@[10,13,5]`	3.000000	3.000000	0	yes
Housing: wall thickness in section z 10	`--section z=10 --wall [0,15,10]`	3.000000	3.000000	0	yes

10 Measurements

M10 Response times

Mouse pass over the hole plate (41 positions, 16 ms each): at most one request in flight, the last position wins. Measured from request until response in the browser (worker, Wasm kernel, SwiftShader without GPU).

- 42 responses, median 2.9 ms, 95 % 48.0 ms

11 Test runs

Run	Result
`cargo test --workspace -- --test-threads=2`	574 passed, 0 failed
`cargo clippy --workspace --all-targets -- -D warnings` (native and `-p cad-wasm --target wasm32-unknown-unknown`) · `cargo fmt --check`	clean · clean
`pnpm typecheck` · `pnpm test` (Vitest)	clean · 233 passed
`pnpm e2e` (Playwright, Chromium, full suite)	131 passed, 0 failed
Tauri E2E T17	passed
Analytic cases via the CLI	21 of 21 within the limit

Flaky tests fixed (previously red sporadically, also without measure mode): `waitHandles` still answered with the handles of the previous cursor line (B06, B10) and now waits for settled handles; `clickAt` in M01–M11 and the click in T17 wait for the click's own effect instead of any response; T17 picks faces with Alt (without point snapping). Nothing skipped or disabled.

New Rust tests (22):

`usage_errors` **ok** · `chamfer_angle_and_at` **ok** · `housing_section_wall_and_distances` **ok** · `usage_errors` **ok** · `hole_plate_radii_bores_fillets_and_pitch` **ok** · `curve::tests::arc_basics` **ok** · `curve::tests::ellipse_length` **ok** · `curve::tests::line_ray` **ok** · `dist::tests::distance_of_cubes` **ok** · `num::tests::quadrature_and_minimum` **ok** · `surface::tests::projections` **ok** · `cylinder_radius_area_length` **ok** · `distances` **ok** · `fillet_and_chamfer` **ok** · `section_through_bores_and_wall` **ok** · `cone_sphere_torus` **ok** · `distance_by_clicks_parallel_planes_and_cli` **ok** · `housing_wall_thickness_in_the_section_and_inner_distance` **ok** · `angle_of_a_chamfer` **ok** · `section_mode_over_the_protocol` **ok** · `snaps_vertices_midpoints_faces_with_provenance` **ok** · `radius_of_bores_and_fillets_in_the_hole_plate` **ok**

cad-measure `tests/analytic.rs`: cylinder (radius, axis, lateral and cap area, circle length, normal distance), cone (radius, half angle, apex, lateral area), sphere and torus (radii, areas, bounding box), fillet cylinder and fillet sphere of a cuboid, fillet torus of a circular edge, chamfer 45°/135°, distances, section through two bores (exact circles, area, wall thicknesses, longitudinal section, oblique section as ellipse); unit tests of curves, surfaces and quadrature. cad-session `tests/measure.rs`: snap (face, vertex, edge midpoint, edge) with provenance, clicks and CLI arguments, references, hole plate, housing with wall thickness in the section, part volume, chamfer, section mode via the protocol. cad-cli `tests/messen.rs`: JSON and text of the new options, usage errors. Vitest `logic/measure.test.ts`.

12 Limitations

- The section needs one kernel Boolean per part and plane; if it fails, there is no section (message), no substitute contour from the mesh.
- Contours on free-form surfaces and on obliquely cut cones, tori and surfaces of revolution remain kernel curves (`exact = false`).
- Wall thickness perpendicular to the contour at one location; the minimum wall thickness of the section is an estimate from samples.
- Distances between free-form surfaces and cases in which the projection ends at a different local minimum hold within the tessellation tolerance; areas of free-form faces from the triangles; volume from the kernel (~10^{-4}).
- CLI references via the nearest point are unambiguous as long as P is closest to one element; results appear in the order of the options.
- 2D results (sketches without extrusion) cannot be measured; the list of measurements is not saved.
- Until the UI rework the context bar sits in the stack at the top right (the view toolbar wraps), the list to the right of the view.
- Tauri T17 operates the view with synthetic pointer events; timings from SwiftShader or software GL are only meaningful relative to each other.